MIL-E-1/1205D
21 MARCH 1972
SUPERSEDING
MIL-E-1/1205C
10 September 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 5BFP7

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus, flat face, dual beam, post accelerator, Mμ metal shield, aluminized screen

DIMENSIONS AND PIN CONNECTIONS: See figures 1 and 2

ABSOLUTE RATINGS:

| Parameter: | Ef | Ec1 | Eb1 | Eb2 | Eb3 | ed |
|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | Vdc | Vdc | v |
| Maximum: | 6.9 | 0, -200 | 1,550 | 4,000 | 15,000 | 600 |
| Minimum: | 5.7 | ------- | ----- | 1,000 | 5,000 | --- |
| TEST CONDITIONS: (see notes 1 and 2) | 6.3 | Adjust | Focus | 2,750 | 13,250 | --- |

ABSOLUTE RATINGS:

| Parameter: | Rg | Zd | Ehk | Eb3/Eb2 | Alt |
|---|---|---|---|---|---|
| Unit: | Meg | Meg (see note 3) | Vdc | Ratio | ft |
| Maximum: | 1.5 | 1.0 | ±180 | 4.9 | 70,000 |
| Minimum: | --- | --- | | --- | |
| TEST CONDITIONS: (see notes 1 and 2) | --- | 1.0 | ---- | 4.82 | ------ |

GENERAL:

Qualification - Required

(D) Revision



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Min | Max | |
| | Qualification | | | | | | | |
| 1027 | Temperature cycling | | --- | --- | -- | 5 | --- | Cycle |
| 1141 | Pressure | | --- | --- | --- | --- | --- | --- |
| 1002 | Barometric pressure, reduced | 33 mm Hg (see note 6) | --- | --- | --- | --- | --- | --- |
| 1026 | Low temperature operation | | --- | --- | --- | -55 | --- | °C |
| 1026 | High temperature operation | | --- | --- | --- | --- | +70 | °C |
| 5111 | Vibration (cathode-ray tubes) | | --- | --- | --- | --- | 1 | mm |
| 1331 | Direct interelectrode capacitance (see note 7) | Cathode to all | --- | --- | C | --- | 6.0 | pF |
| | | Grid 1 to all | --- | --- | C | --- | 6.5 | pF |
| | | D1 to D2 | --- | --- | C | --- | 4.0 | pF |
| | | D3 to D4 | --- | --- | C | --- | 3.5 | pF |
| | | D1 to all | --- | --- | C | --- | 17.0 | pF |
| | | D2 to all | --- | --- | C | --- | 19.0 | pF |
| | | D3 to all | --- | --- | C | --- | 15.5 | pF |
| | | D4 to all | --- | --- | C | --- | 15.5 | pF |
| 5101 | Neck and bulb alinement (electrostatic types) | | --- | --- | --- | --- | ±1/16 | inch |
| 5101 | Face tilt | | --- | --- | --- | --- | 0.080 | inch |
| 5248 | Deflection-factor uniformity | | --- | --- | --- | --- | --- | --- |
| 5256 | Magnetization | | --- | --- | --- | --- | --- | --- |
| 5250 | Interaction factor | | --- | --- | --- | --- | $14 \times 10^{-6}$ | In/Vdc |
| ---- | Shock | See note 8 | --- | --- | --- | --- | --- | --- |
| 1216 | Base material insulating quality | Zone 5 (minimum) | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 1 | | | | | | | |
| ---- | Marking | See note 4 | --- | --- | --- | --- | --- | --- |
| 5106 | Bulb, screen, and faceplate quality | | --- | --- | --- | --- | --- | --- |
| 5201 | Electrode currents (anode No. 3) | See note 10 | --- | --- | --- | --- | 65 | µAdc |
| 5201 | Voltage breakdown | | --- | --- | --- | --- | --- | --- |
| 5201 | Voltage breakdown (electrostatic types) | Shield connected to anode 2 | --- | --- | --- | --- | --- | --- |
| 5206 | Gas "cross" | Ib3 = 50 µAdc See note 15 | --- | --- | --- | --- | --- | --- |
| ---- | Trace alinement | See note 5 | --- | --- | --- | --- | --- | --- |
| 5101 | Angle between traces | | --- | --- | --- | 88 | 92 | Degree |
| 5101 | Angle between corresponding traces | | --- | --- | --- | --- | 2 | Degree |



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Min | Max | |
| | Quality conformance inspection, part 1 -Continued | | | | | | | |
| 5223 | Modulation | Ib3 = 50 µAdc (see note 11) | --- | --- | Δ Ec1 | --- | 36 | Vdc |
| 5226 | Line width A (electrostatic deflection) | Ib3 = 50 µAdc | --- | --- | --- | --- | 0.5 | mm |
| 5226 | Line width B | Ib3 = 50 µAdc (see note 12) | --- | --- | --- | --- | 0.6 | mm |
| 5231 | Spot position (electrostatic deflection) | | --- | --- | --- | --- | 16 | mm |
| 5251 | Spot displacement (leakage) | | --- | --- | --- | --- | 10 | mm |
| 5241 | Grid cutoff voltage | | --- | --- | Ec1 | -65 | -35 | Vdc |
| 5246 | Focusing voltage at cutoff | | --- | -- | Eb1 | --- | 900 | Vdc |
| 5248 | Deflection factor | 1D2 | --- | --- | DF | 130 | 161 | Vdc/in. |
| 5248 | Deflection factor | 3D4 | --- | --- | DF | 110 | 138 | Vdc/in. |
| 5251 | Grid No. 1 leakage | | --- | --- | --- | --- | 3 | µAdc |
| 5251 | Anode No. 1 leakage | | --- | --- | --- | --- | 5 | µAdc |
| 5251 | Anode No. 2 leakage | | --- | --- | --- | --- | 5 | µAdc |
| ---- | Focusing voltage | Ib3 = 0-50 µAdc | --- | --- | --- | 700 | 900 | Vdc |
| ---- | Useful scan | See note 13 | --- | --- | --- | --- | --- | --- |
| ---- | Pattern distortion | See note 9 | ---- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1101 | Secureness of base, base insert, and cap | | --- | --- | --- | --- | --- | --- |
| 1301 | Heater current | See note 14 | --- | --- | If | 540 | 660 | mA |
| 5201 | Electrode currents (anode No. 1) | | --- | --- | --- | -10 | +5 | µAdc |
| 5201 | Electrode currents (cathode) | Ib3 = 100 µAdc | --- | --- | Ik | --- | 1 | mAdc |
| 5201 | Electrode currents (beam) | Ec1 = 0 | --- | --- | Ib3 | 100 | --- | µAdc |
| 5101 | Base alinement (electrostatic types) | +3D4, pin 4 | --- | --- | --- | --- | --- | --- |
| 5101 | Side terminal alinement (electrostatic types) | +3D4 | --- | --- | --- | --- | --- | --- |
| 5101 | Side terminal and base alinement | Pin | --- | --- | --- | --- | --- | --- |
| (D)1111 | Base pin solder depth | | --- | --- | --- | --- | --- | --- |
| 5101 | Neck and base alinement (electrostatic types) | | --- | --- | --- | --- | --- | --- |
| 5216 | Cathode illumination | | --- | --- | --- | --- | $10 \times 10^{-6}$ | ftL |



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| METHOD | TEST OR REQUIREMENT | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Min | Max | |
| | Quality conformance inspection, part 2 | | | | | | | |
| 5216 | Stray emission (conventional types) | Eb2 = 4,000 Vdc; Eb3 = 15,000 Vdc | --- | --- | --- | --- | --- | --- |
| 5221 | Screens | | --- | --- | --- | --- | --- | --- |
| 5246 | Focusing voltage at modulation condition | | --- | --- | Eb1 | 600 | 900 | Vdc |
| 5251 | Heater-cathode leakage | | --- | --- | Ihk | --- | 15 | µAdc |
| 1105 | Permanence of marking | | --- | --- | --- | --- | --- | --- |
| 5216 | Grid emission | | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | |
| ---- | Life-test provisions | Group C<br>Ib3 = 12 µAdc;<br>t = 500 hours (min)<br>Eb2 = 4,000 Vdc;<br>Eb3 = 15,000 Vdc | --- | --- | --- | --- | --- | --- |
| ---- | Life-test end points: | Ib3 = 37.5 µAdc | --- | --- | --- | --- | --- | -- |
| | Line width A | | --- | --- | --- | --- | 0.5 | mm |
| | Line width B | | --- | --- | --- | --- | 0.6 | mm |
| | Modulation | | --- | --- | Δ Ec1 | --- | 36 | Vdc |
| | Grid 1 leakage | | --- | --- | --- | --- | --- | --- |
| | Anode 1 leakage | | --- | --- | --- | --- | --- | --- |
| | Anode 2 leakage | | --- | --- | --- | --- | --- | --- |
| | Heater cathode leakage | | --- | --- | --- | --- | --- | --- |
| | Stray emission | | --- | --- | --- | --- | --- | --- |



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NOTES:

1. All readings of beam current (Ib3) shall be in addition to the reading obtained for method 5221 anode 3 current.

2. All tests except capacitance, vibration, interaction factor, and environmental tests shall be made on each unit separately. Each deflection plate shall be connected to A2 through a resistance of 1.0 megohm.

3. It is recommended that the deflecting electrode circuit resistances be approximately equal. Higher resistance values up to 5.0 megohms may be used for low beam current operation.

4. In addition to the requirements of 3.6, marks shall be placed on the tube face as shown on figure 4 for the purpose of orienting the tube in its associated equipment. These marks shall indicate the planes defined by the vertical trace of gun A and the horizontal trace of gun B. The marks shall be removable by the application of a common solvent such as acetone, carbon tetrachloride, or water. The marks shall not be removable by gentle rubbing.

5. The 1D2 (horizontal) trace of gun B shall be 90° ±2° from the 3D4 (vertical) trace of gun A. The plane through the axis of both guns shall be considered horizontal.

6. For low pressure test on anode 3, place metal foil just touching the high voltage potting compound on all sides as shown on figure 5, and in direct contact with the glass for a distance of at least a half inch in all directions from the base of the potting compound. The metal foil shall be grounded. The test shall be conducted in accordance with the specified test conditions and altitude. The tubes which exhibit arcing or corona during test shall be considered defective.

7. All other electrodes contained within the envelope and the Mμ metal shield shall be tied together.

8. The test shall be performed in accordance with method 205, test condition A, of MIL-STD-202.

   a. The following conditions apply:

      (1) The tube shall be mounted to the elevator table by a suitable rigid assembly. Hardwood clamps shall be used at the face plate and near the neck to secure the tubes to the assembly. Hardwood or metal clamps may be used at similarly appropriate points on tubes having integral shields.

      (2) The assembly shall be such as to permit changes in tube orientation. A total of 10 shocks shall be applied to the tube in the following sequencies

| Order | Number of drops | Direction |
|---|---|---|
| 1 | 2 | Base to face |
| 2 | 2 | x+ |
| 3 | 2 | x- |
| 4 | 2 | y+ |
| 5 | 2 | y- |

      (3) No potentials shall be applied during shock.

   b. After shock test, cathode-ray tubes shall meet the requirements specified and shall be subjected to the following tests:

      (1) The change in position of the undeflected focused spot from its position noted before the test (see method 5231) shall not exceed 4 mm.



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NOTES: -Continued

(2) The change in deflection factor for electrostatic tube types shall not exceed 5 percent (see method 5248).

(3) Starting with a tube with no loose particles, the maximum number of loose particles as a result of shocking shall be no more than a total of five loose particles greater than 0.016 inch in any one cathode-ray tube. Tubes having loose particles greater than 0.016 inch shall be vibrated with the face down in a horizontal plate at 5G's for 1 minute. If the screen shows damage from such vibration, the tube shall be rejected.

(4) The change in grid cutoff voltage shall not exceed 10 percent (see method 5241).

(5) There shall be no damage to the external parts of the tube.

9. The total vertical movement of the upper end of a 3-inch vertical trace (centered with respect to the tube face), deflected horizontally 1.5 inches to the left and to the right of the center of the tube face, shall not exceed 0.10 inch. The 3D4 trace shall be considered vertical. The sum of the total horizontal movement of the left and right ends of a 3-inch horizontal trace (centered with respect to the tube face), deflected vertically 1.5 inches above and below the center of the tube face, shall not exceed 0.10 inch. The 1D2 trace shall be considered horizontal.

10. With both beams cutoff, the post-accelerator anode (A3) will draw 65 µAdc maximum.

11. Do not exceed zero grid bias.

12. Line width B shall be measured at ±1.625 inches from the tube face center.

13. Useful scan display on Face of the tube shall be as shown in figure 6.

14. There are two independent heaters.

15. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy - EC
Air Force - 11, 17

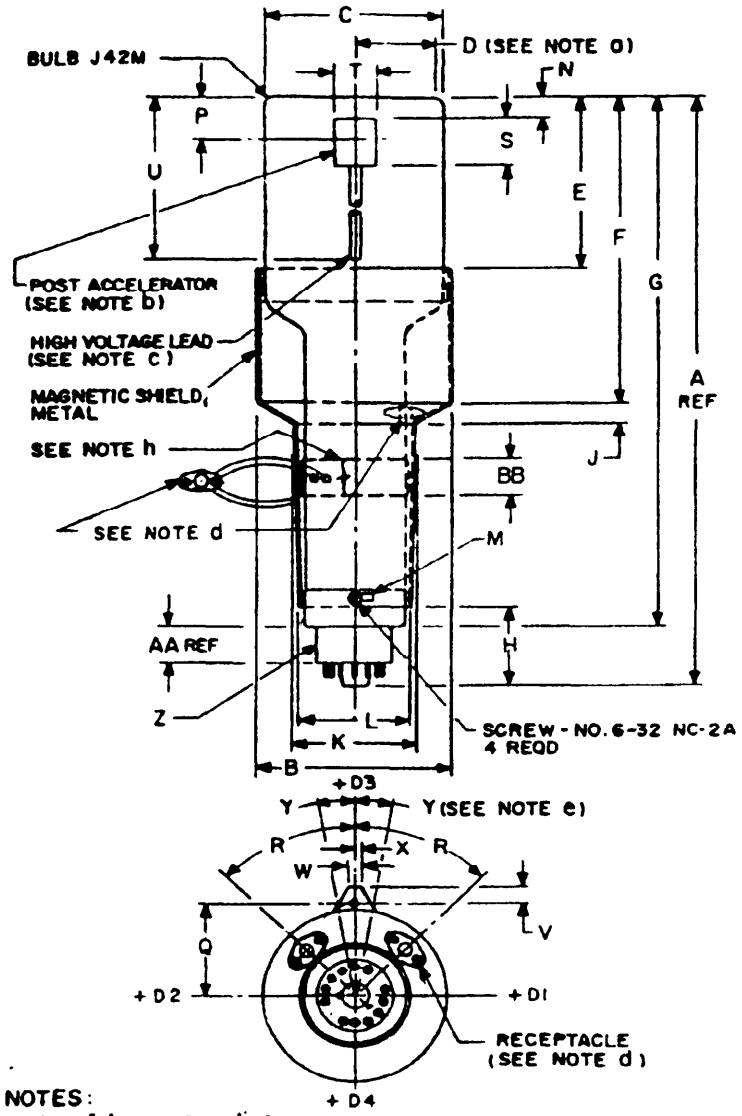
User activities:
Army - NONE
Navy - MC, CG
Air Force - NONE

Preparing activity:
Air Force - 80

(Project 5960-2526)



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| DIMENSIONS | | | | |
|---|---|---|---|---|
| LTR | INCHES | | MILLIMETERS | |
| | MIN | MAX | MIN | MAX |
| QUALIFICATION | | | | |
| Z | BASE: B12-231 (SEE FIGURE 2) | | | |
| QUALITY CONFORMANCE INSPECT, PART 2 | | | | |
| A | | 18.38 | | 466.85 |
| B | | 5.41 | | 137.41 |
| C | 5.16 | 5.34 | 131.06 | 135.64 |
| D | 2.25 | | 57.15 | |
| E | 5.25 | 5.75 | 133.35 | 146.05 |
| F | | 9.50 | | 241.3 |
| G | 16.47 | 16.53 | 418.34 | 419.86 |
| H | 2.44 | 2.56 | 61.98 | 65.02 |
| J | .57 | .67 | 14.48 | 17.02 |
| K | | 3.41 | | 86.61 |
| L | | 3.25 | | 82.55 |
| N | .625 | | 15.88 | |
| P | 1.250 | 1.625 | 31.75 | 41.28 |
| Q | | 2.93 | | 74.42 |
| R | 40° | 50° | | |
| S | | 3.00 | | 76.2 |
| T | | 1.30 | | 33.02 |
| U | 12.5 | 13.5 | 317.5 | 342.9 |
| V | | .50 | | 12.7 |
| W | | .94 | | 23.88 |
| X | | .47 | | 11.94 |
| Y | | 10° | | |
| BB | 1.0 | | 25.4 | |
| AA | 1.050 | 1.110 | 26.67 | 28.19 |
| REFERENCE DIMENSIONS (NOTE f) | | | | |
| M | 4 PCS. 90° APART .5 NOM (12.7 MM) | | | |

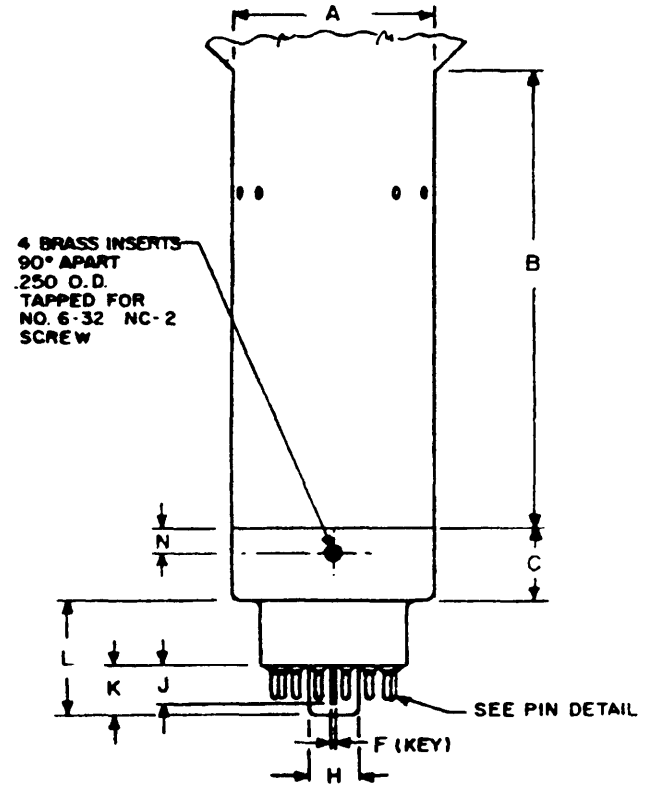
NOTES:

a. Useful screen radius.
b. Post accelerator contact shall be located on side of bulb which corresponds with pin NO. 4 and shall be corona free. Contact and corresponding termination of high voltage lead shall be imbedded in a nonconducting material which shall be bonded to the lead insulation and to the glass bulb of the tube. Bulb contact cover shall be molded of nonconducting material. Configuration of bulb contact cover is optional with the provision that dimensions shown are not exceeded.
c. High voltage lead insulation shall be flexible and shall withstand low pressure test. Free end of lead shall terminate in female contact connector and nonconducting coupling ring which will mate with high voltage male contact connector shown on figure 7 and which will grip the lead insulation. Female contact connector shall be Amp, type 832-692, or equivalent.
d. Two receptacles shall be Winchester type M9S or equivalent.
e. Allowable deviation of connector from true position on centerline.
f. Reference dimensions are provided for information only and are not required for inspection purposes.
g. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
h. The deflection plate leads emerging thru the neck of the tube shall be insulated with a material having a dielectric strength of 500 Vdc.
j. Unless otherwise specified, AQL (percent defective) of 6.5 and Inspection level of S3 applies.

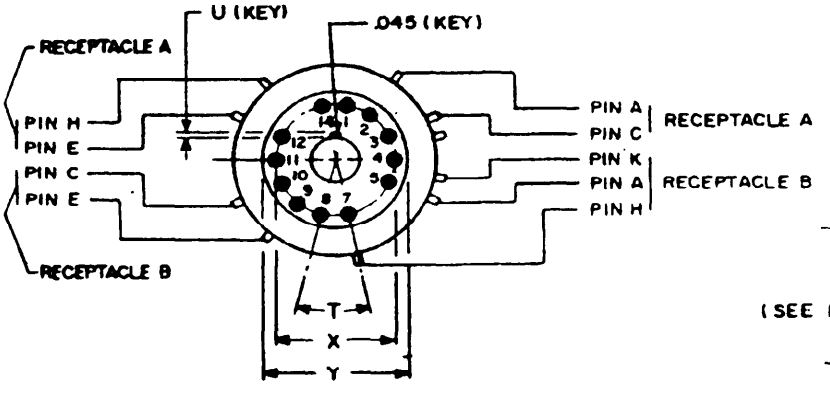
FIGURE 1. Outline drawing of electron tube 5BFP7

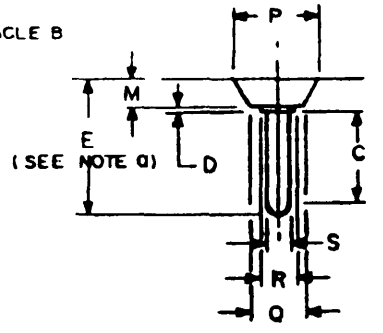


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| DIMENSIONS | | | | |
|---|---|---|---|---|
| LTR | INCHES | | MILLIMETERS | |
| | MIN | MAX | MIN | MAX |
| A | 2.955 | 3.035 | 75.06 | 77.09 |
| B | 8.21 | 8.31 | 208.53 | 211.07 |
| C | 1.155 | 1.175 | 29.34 | 29.85 |
| D | | .010 | | .25 |
| E | | .515 | | 13.08 |
| F | .075 | .095 | 1.91 | 2.41 |
| G | .340 | | 8.64 | |
| H | .740 | .765 | 18.80 | 19.43 |
| J | .615 | | 15.62 | |
| K | | .765 | | 19.43 |
| L | 1.825 | 1.845 | 46.36 | 46.86 |
| M | | .110 | | 2.79 |
| N | .334 | .354 | 8.48 | 8.99 |
| P | .315 | .325 | 8.00 | 8.26 |
| Q | .195 | .205 | 4.95 | 5.21 |
| R | | .135 | | 3.43 |
| S | .090 | .096 | 2.29 | 2.44 |
| T | | 25-5/7° | | |
| U | .073 | .083 | 1.85 | 2.11 |
| X | 1.740 | 1.760 | 44.20 | 44.70 |
| Y | 2.179 | 2.259 | 55.35 | 57.38 |



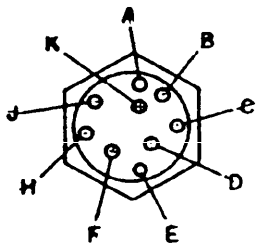

PIN DETAIL

a. On finished tube add .030 (.76 mm) for solder.
b. Pin diameters do not include increase due to solder.
c. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

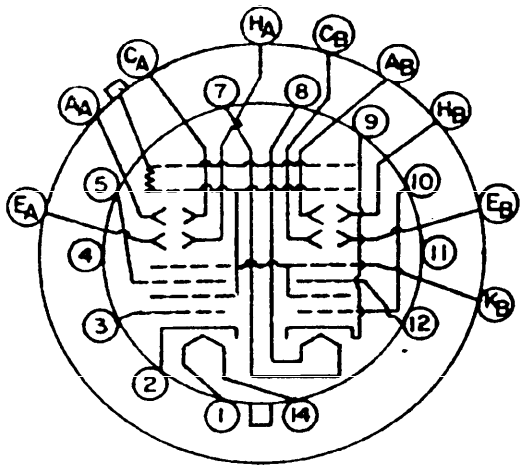
FIGURE 2. Base B12-231



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WINCHESTER M9S CONNECTOR
OR EQUIVALENT.




BOTTOM VIEW

| UNIT-A | | UNIT-A | |
|---|---|---|---|
| ELEMENT | RECEPTACLE | PIN NO. | ELEMENT |
| Deflecting electrode D1 | C | 1 | Heater |
| Deflecting electrode D2 | A | 2 | Cathode |
| Deflecting electrode D3 | H | 3 | Grid No. 1 |
| Deflecting electrode D4 | E | 5 | Focusing electrode |
| Shield | B | 14 | Heater |
| UNIT-B | | UNIT-B | |
| Accelerator | K | 7 | Heater |
| Deflecting electrode D1 | H | 8 | Heater |
| Deflecting electrode D2 | A | 9 | Cathode |
| Deflecting electrode D3 | E | 10 | Grid No. 1 |
| Deflecting electrode D4 | C | 12 | Focusing electrode |

FIGURE 3. Base diagram of electron tube type 5BFP7



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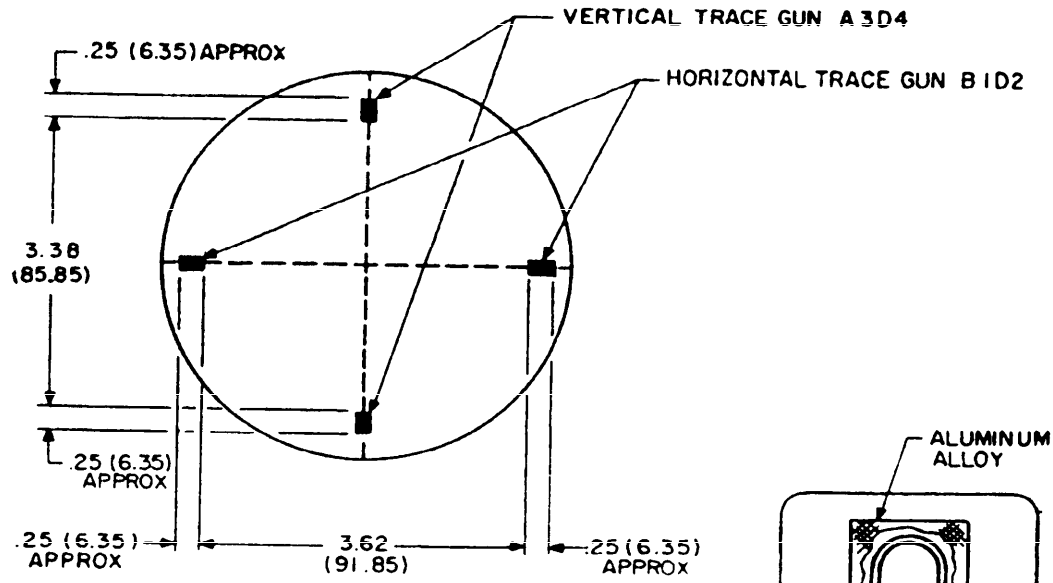



FIGURE 4. Trace orientation marks

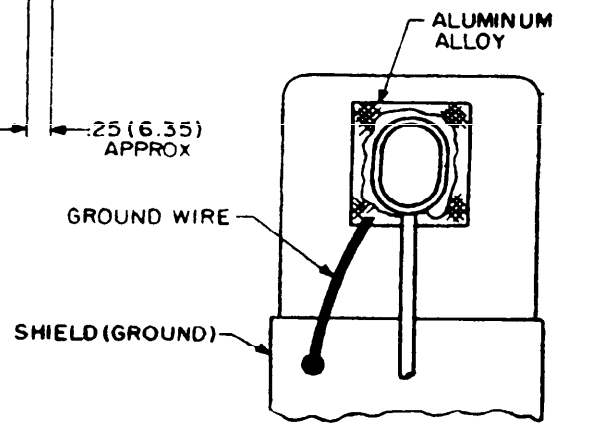



FIGURE 5. Arrangement for low pressure test

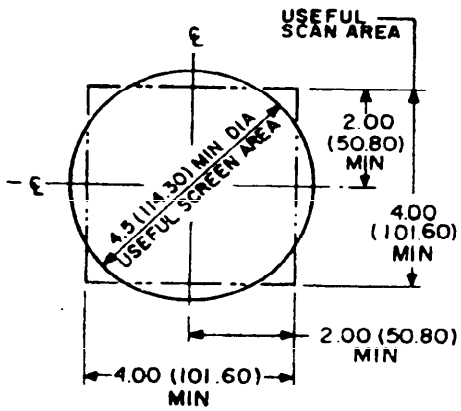



FIGURE 6. Useful scan area

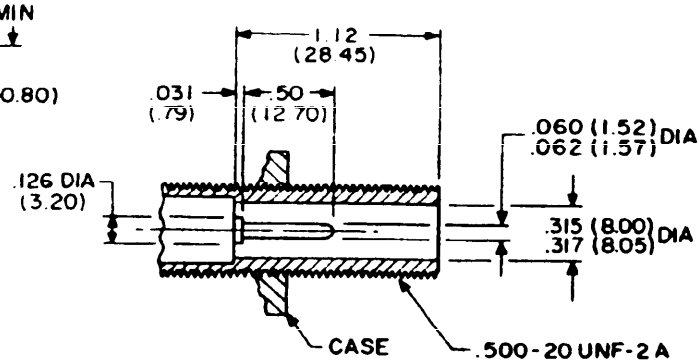



FIGURE 7. High voltage connector

NOTE :
Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm . (millimeters shown in parentheses.)



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

**DD FORM 1 JAN 66 1426** REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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POSTAGE AND FEES PAID
Department of the Air Force

OFFICIAL BUSINESS

SMAMA (MMSSC)
MC CLELLAN AFB, CALIF. 95652

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